

Splunk Journey – Day 02

Date: 2026-06-11

Objective

Install and configure the BOTS v3 dataset in a local Splunk Enterprise lab to support SOC analyst training, threat hunting, and SPL practice. Additionally, explore fundamental Splunk concepts.

Dataset Acquisition

The official BOTS v3 dataset was downloaded from the Splunk GitHub repository.

- Repository: <https://github.com/splunk/botsv3/>
- Download: wget https://botsdataset.s3.amazonaws.com/botsv3/botsv3_data_set.tgz

```
debian@debian:~$ wget https://botsdataset.s3.amazonaws.com/botsv3/botsv3_data_set.tgz
--2026-06-10 23:33:39-- https://botsdataset.s3.amazonaws.com/botsv3/botsv3_data_set.tgz
Resolving botsdataset.s3.amazonaws.com (botsdataset.s3.amazonaws.com)... 52.217.224.49, 52.216
.217.121, 16.182.35.57, ...
Connecting to botsdataset.s3.amazonaws.com (botsdataset.s3.amazonaws.com)|52.217.224.49|:443..
. connected.
HTTP request sent, awaiting response... 200 OK
Length: 335251397 (320M) [application/gzip]
Saving to: 'botsv3_data_set.tgz'

botsv3_data_set.tgz      100%[=====>] 319.72M  9.59MB/s   in 48s

2026-06-10 23:34:29 (6.66 MB/s) - 'botsv3_data_set.tgz' saved [335251397/335251397]
```

Integrity verification:

```
debian@debian:~$ md5sum botsv3_data_set.tgz
d7ccca99a01cff070dff3c139cdc10eb  botsv3_data_set.tgz
```

```
p1.py x s.py x
1 s1 = "d7ccca99a01cff070dff3c139cdc10eb"
2 s2 = "d7ccca99a01cff070dff3c139cdc10eb"
3
4 if(s1==s2):
5     print("md5 hash verify")
6 else:
7     print("not")
```

```
md5 hash verify
[Finished in 107ms]
```

Configuration

1. Extract the downloaded dataset and copy it to the appropriate directory.

```
debian@debian:~/botstv3_data_set$ sudo cp -r ~/botstv3_data_set /opt/splunk/etc/apps/
debian@debian:~/botstv3_data_set$ ls /opt/splunk/etc/apps/
alert_logevent      sample_app          splunk_instrumentation
alert_webhook       search             splunk_internal_metrics
appsbrowser         splunk-ai-canvas   SplunkLightForwarder
audit_trail         splunk_app_for_splunk_olly_cloud splunk_metrics_workspace
botstv3_data_set    splunk_archiver    splunk_monitoring_console
introspection_generator_addon splunk-dashbaord-studio splunk_pipeline_builders
journald_input      splunk-data-management splunk_rapid_diag
launcher            SplunkDeploymentServerConfig splunk-rolling-upgrade
learned             SplunkForwarder    splunk_secure_gateway
legacy              splunk_gdi         splunk-visual-exporter
python_upgrade_readiness_app splunk_httpinput   user-prefs
```

2. Set the required permissions for the directory.
3. Restart the Splunk service.

```

debian@debian:~/botstv3_data_set$ sudo chown -R splunk:splunk /opt/splunk/etc/apps/botstv3_data_set
debian@debian:~/botstv3_data_set$ ls -ld /opt/splunk/etc/apps/botstv3_data_set
drwxr-xr-x 6 splunk splunk 4096 Jun 11 01:33 /opt/splunk/etc/apps/botstv3_data_set
debian@debian:~/botstv3_data_set$ sudo -u splunk splunk restart

```

Check

The screenshot displays the Splunk Enterprise web interface. At the top, the search bar contains the query `index=botstv3 earliest=0`. Below the search bar, the results are displayed in a table format. The first event is an error message from Splunkd at 9/19/19 2:10:50.000 PM. The second event is a network packet capture at 8/20/18 11:27:09.296 AM, showing details like bytes, destination IP, and protocol.

Now learn about some basic topics in splunk. Such as,

1. Using splunk web
2. Using search
3. Exploring events
4. Using search terms

Using splunk web

There are three predefined default roles in splunk Enterprise:

1. Administrator
2. Power User
3. User

Search & Reporting app

The screenshot shows the Splunk Search & Reporting app interface. At the top, the search bar contains the query `index=botsv3 earliest=0`. Below the search bar, it indicates that 47,884 of 39,991 events matched. The interface includes a sidebar with field discovery, a search bar, and a main results pane with a table of events. The table shows two events: one from 9/19/19 and another from 8/20/18. The 8/20/18 event is expanded, showing detailed network and system information.

Time	Event
9/19/19 2:10:50.000 PM	<error><message>Splunkd daemon is not responding: ('Error connecting to /services/shcluster/config/config: [Errno 111] Connection refused').</message></error> > host = OD-FM-NA-10ad20665d4bdace22.amazonaws.com source = ess_content_importer sourcetype = ess_content_importer
8/20/18 11:27:09.296 AM	([-] bytes: 58637 bytes_in: 38 bytes_out: 58687 dest_ip: 172.16.0.178 dest_mac: 02:48:38:18:83:78 endtime: 2018-08-20T15:27:09.296788Z flow_id: c240fdab-997d-4285-a221-1080976a6b8a fragment_count: 0 packets_in: 1 packets_out: 36 protocol: UDP protocol_stack: ip:udp:unknown protoid: 17 src_ip: 13.125.33.130 src_mac: 02:4f:07:61:53:04 Timestamp: 2018-08-20T15:27:09.296118Z

- ❖ [Note] Limiting a search by time is key to faster results and is a best practice
- ❖ Commands that create statistics and visualizations are called transforming commands.
- ❖ By default, a search job will remain active for 10 minutes
- ❖ Shared search jobs remain active for 7 days.

The screenshot shows the Splunk Search & Reporting app interface with a 'Share Job' dialog box open. The dialog box contains the message: 'The job's lifetime has been extended to 7 days and read permissions set to everyone. Manage the job'. Below this message is a 'Copy job link' button. The background search results show 2,083,056 events. The interface includes a sidebar with field discovery, a search bar, and a main results pane with a table of events.

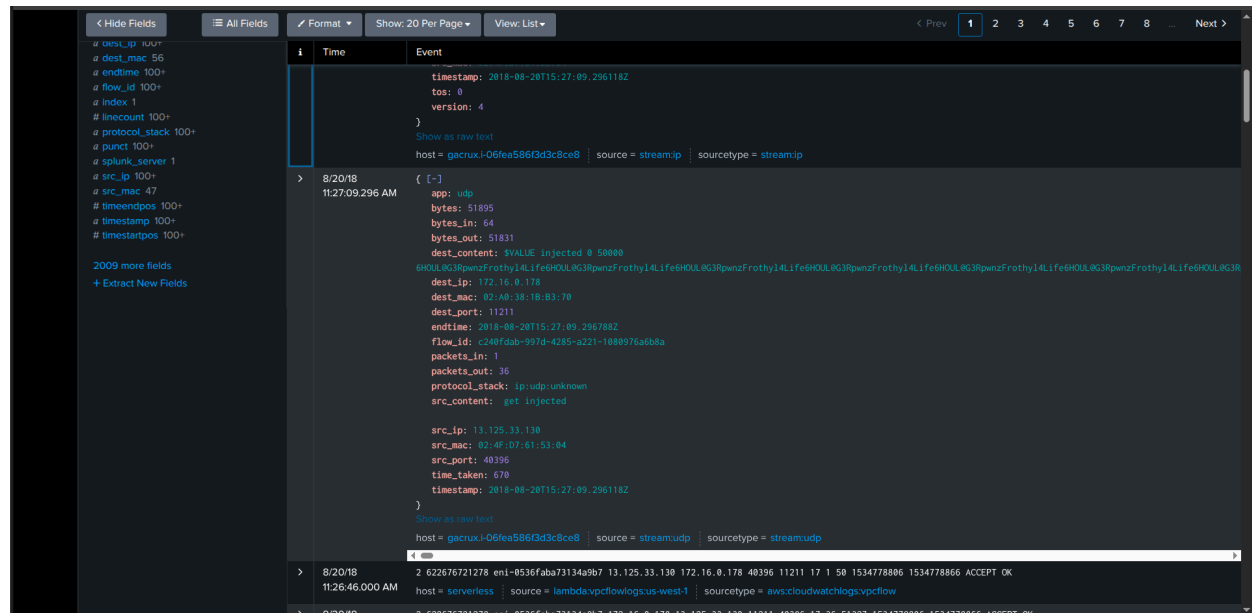
Time	Event
9/19/19 2:10:50.000 PM	<error><message>Splunkd daemon is not responding: ('Error connecting to /services/shcluster/config/config: [Errno 111] Connection refused').</message></error> > host = OD-FM-NA-10ad20665d4bdace22.amazonaws.com source = ess_content_importer sourcetype = ess_content_importer
8/20/18 11:27:09.296 AM	([-] bytes: 58637 bytes_in: 38 bytes_out: 58687 dest_ip: 172.16.0.178 dest_mac: 02:48:38:18:83:78 endtime: 2018-08-20T15:27:09.296788Z flow_id: c240fdab-997d-4285-a221-1080976a6b8a fragment_count: 0 packets_in: 1 packets_out: 36 protocol: UDP protocol_stack: ip:udp:unknown protoid: 17 src_ip: 13.125.33.130 src_mac: 02:4f:07:61:53:04 Timestamp: 2018-08-20T15:27:09.296118Z

- ❖ Field discovery is disabled in fast mode

Exploring Events:

Showing the newest events first.

Default selected field —> host, source, sourcetype



Using search terms:

Search terms are not case sensitive

Wildcard ———> fail* (* asterisk)

Booleans ———> and, or , not

If no boolean is used, AND is implied

Boolean operation have an order of evaluation

1. NOT
2. OR
3. AND

Parentheses can be used to control the order of evaluation

Example:

Failed NOT (Success OR accepted)

Adding parentheses around the OR statement in the search terms, splunk will first evaluate the terms in the parentheses.

Exact phrases can be searched by placing the terms in quotes.

Example:

“Failed password”

Use a backslash character to escape quotes.

Example:

ifo="user \"chrsv4\" not in database"